

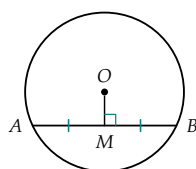
Circle Geometry

Every theorem, its converse, and the exact reason to write beside it

Pills like $\angle$ in alt. segment are the **HKEAA-style reason abbreviations**: write them in brackets after each step. Throughout, O is the centre.

I Chords perpendiculars from the centre

The chord trio



Any two of {through O , $\perp AB$, bisects AB } give the third.

$OM \perp AB \Rightarrow AM = MB$

$\perp$ from centre bisects chord

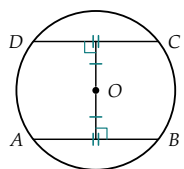
$AM = MB \Rightarrow OM \perp AB$

line from centre to mid-pt. of chord $\perp$ chord

$\perp$ bisector of AB passes through O

$\perp$ bisector of chord passes through centre

Equal chords $\Leftrightarrow$ equidistant



$AB = CD \Rightarrow OM = ON$

equal chords, equidistant from centre

$OM = ON \Rightarrow AB = CD$

chords equidistant from centre are equal

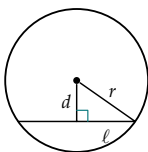
Distance to a chord always means the perpendicular distance.

Chord length & the magic right triangle

Radius r , chord length 2ℓ , distance d from O :

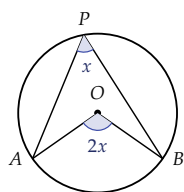
$$r^2 = d^2 + \ell^2 \quad \text{chord} = 2\sqrt{r^2 - d^2}$$

Whenever a chord and a radius meet, drop the perpendicular from O and use Pythagoras.



II Angles in a circle centre, circumference, segment

Angle at centre

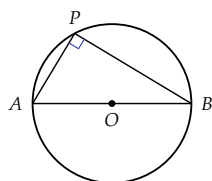


$\angle AOB = 2\angle APB$

$\angle$ at centre twice $\angle$ at $\odot$ ce

Both angles stand on the **same arc** AB . Also true when $\angle AOB$ is *reflex* (then P lies on the minor arc) and in the "arrowhead" case where O is outside $\triangle APB$.

Angle in a semicircle



AB diameter $\Rightarrow \angle APB = 90^\circ$

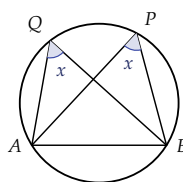
$\angle$ in semi-circle

$\angle APB = 90^\circ \Rightarrow AB$ is a diameter

converse of $\angle$ in semi-circle

Special case of the centre theorem ($2x = 180^\circ$).

Angles in the same segment



$\angle APB = \angle AQB$

$\angle$ s in the same segment

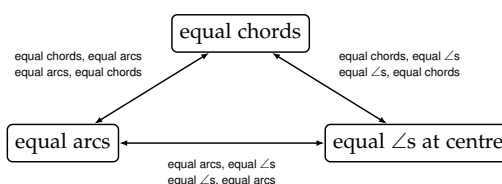
P, Q must be on the **same side** of AB .

Opposite sides $\Rightarrow$ the angles are *supplementary* instead (cyclic quad).

Converse: if $\angle APB = \angle AQB$ (same side), then A, B, P, Q concyclic

converse of $\angle$ s in the same segment

Chords, arcs and angles move together



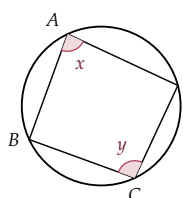
Proportionality (arcs, not chords!): $\frac{\text{arc } AB}{\text{arc } CD} = \frac{\angle AOB}{\angle COD} = \frac{\angle APB}{\angle CQD}$

arcs prop. to $\angle$ s at centre arcs prop. to $\angle$ s at $\odot$ ce

Chord length is *not* proportional to angle: doubling the angle does not double the chord. Whole circle = 360° at centre = 180° at circumference.

III Cyclic quadrilaterals four points on one circle

Opposite angles



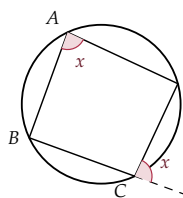
$x + y = 180^\circ$ (and $\angle B + \angle D = 180^\circ$)

opp. $\angle$ s, cyclic quad.

Converse: if one pair of opposite angles is supplementary, $ABCD$ is cyclic

opp. $\angle$ s supp.

Exterior angle



$\angle DCE = \angle BAD$ (ext. $\angle$ = interior opposite $\angle$)

ext. $\angle$, cyclic quad.

Converse: ext. $\angle$ = int. opp. $\angle \Rightarrow$ cyclic

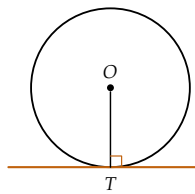
ext. $\angle$ = int. opp. $\angle$

Match with the vertex *diagonally opposite*, not the adjacent one.

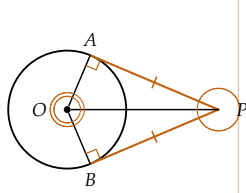
Proving four points are concyclic: the three tests

- 1 a pair of opposite angles sums to 180° $\text{opp. } \angle\text{s supp.}$
- 2 an exterior angle equals the interior opposite angle $\text{ext. } \angle = \text{int. opp. } \angle$
- 3 a side subtends equal angles on the same side $\text{converse of } \angle\text{s in the same segment}$

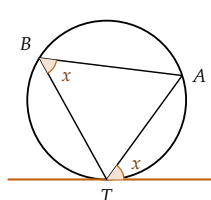
Special case: two right angles on the same segment $\Rightarrow$ concyclic on that segment as diameter ($\text{converse of } \angle \text{ in semi-circle}$). Every triangle is cyclic; a parallelogram is cyclic *only* if it is a rectangle; an isosceles trapezium is always cyclic.

IV Tangents one line, one touching point**Tangent $\perp$ radius**

ST tangent at $T \Rightarrow OT \perp ST$
 $\text{tangent } \perp \text{ radius}$
 $OT \perp ST$ at T on the circle $\Rightarrow ST$ is a tangent
 $\text{converse of tangent } \perp \text{ radius}$
Reflex: tangent at $T \Rightarrow$ join OT immediately.

Tangents from an external point

PA, PB tangents from P :
 $PA = PB$
 $\angle AOP = \angle BOP$
 $\angle APO = \angle BPO$
 $\text{tangent properties}$
 Here $\triangle OAP \cong \triangle OBP$ (RHS), and $OAPB$ is a kite with two right angles $\Rightarrow$ cyclic, with $\angle AOB + \angle APB = 180^\circ$.

Angle in the alternate segment

$\angle ATS = \angle TBA$
 $\angle \text{ in alt. segment}$
 Angle between tangent and chord = angle on the chord in the segment on the *other side* of the chord.
 Converse: if $\angle ATS = \angle TBA$ then ST is tangent
 $\text{converse of } \angle \text{ in alt. segment}$

How to spot the alternate segment. Stand at T . The tangent-chord angle $\angle ATS$ "opens onto" the arc TA between the chord and S . The equal angle sits at any point B on the *far* arc, looking back at chord TA . The angle on the other side, $\angle ATS'$, equals the angle at a point on the *near* arc.

V Centres of a triangle four lines, four points**Where the lines meet**

centre	lines	defining property
circumcentre	$\perp$ bisectors of sides	equidistant from the 3 <i>vertices</i> ; centre of circumcircle
in-centre	$\angle$ bisectors	equidistant from the 3 <i>sides</i> ; centre of incircle; always inside
centroid	medians	divides each median 2 : 1 from vertex; always inside; $G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$
orthocentre	altitudes	where the three heights meet

Right-angled $\triangle$: circumcentre = mid-pt. of hypotenuse ($\angle$ in semi-circle); orthocentre = the right-angled vertex. **Obtuse:** circumcentre and orthocentre lie *outside*. **Equilateral:** all four coincide. **Isosceles:** all four lie on the axis of symmetry.

VI Circles on the coordinate plane equations**Equations of a circle**

Standard: $(x-h)^2 + (y-k)^2 = r^2$, centre (h,k) , radius r .

General: $x^2 + y^2 + Dx + Ey + F = 0$

$$\text{centre} = \left(-\frac{D}{2}, -\frac{E}{2} \right), \quad r = \sqrt{\left(\frac{D}{2} \right)^2 + \left(\frac{E}{2} \right)^2 - F}.$$

Real circle needs $\left(\frac{D}{2} \right)^2 + \left(\frac{E}{2} \right)^2 - F > 0$. Coefficients of x^2 and y^2 must be **equal** (divide through first) and there is **no** xy term.

Point (x_0, y_0) vs circle: put it into LHS $= x_0^2 + y_0^2 + Dx_0 + Ey_0 + F$: < 0 inside, $= 0$ on, > 0 outside. (Or compare its distance to the centre with r .)

Lines meeting circles

Substitute the line into the circle $\rightarrow$ quadratic $ax^2 + bx + c = 0$, discriminant Δ :

$$\begin{array}{ccc} \Delta > 0 & \Delta = 0 & \Delta < 0 \\ 2 \text{ intersections} & \text{tangent} & \text{no intersection} \end{array}$$

Mid-point of chord: $x_M = \frac{x_1 + x_2}{2} = -\frac{b}{2a}$ (sum of roots), so there is no need to solve.

Tangent at P on circle, centre C : slope $= -\frac{1}{m_{CP}}$ (tangent $\perp$ radius).

Tangent with given slope m : write $y = mx + c$, set $\Delta = 0$, solve for c (two answers).

Chord length: find the distance d from centre to line, then $2\sqrt{r^2 - d^2}$.

Circle through 3 points: substitute each into the general form and solve for D, E, F , or find the circumcentre as the meet of two $\perp$ bisectors.

Line from centre to mid-pt. of a chord is $\perp$ to it:

$$m_{CM} \cdot m_{\text{chord}} = -1.$$

VII Proof craft how the marks are won

Reflexes: see this → do this

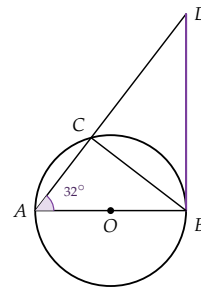
tangent at T	join OT (tangent $\perp$ radius)
tangent + chord at T	look for $\angle$ in alt. segment
two tangents from P	(tangent properties); $\triangle PAB$ isosceles
diameter	look for 90° ($\angle$ in semi-circle)
two radii OA, OB	isosceles $\triangle OAB$ (base $\angle$ s, isos. $\triangle$)
angle at centre	find an angle at $\odot$ ce on the same arc
4 points on circle	opp. angles & ext. angle
chord & centre	drop $\perp$ from O ; Pythagoras
equal arcs/chords	equal angles, and vice versa
"prove concyclic"	one of the three tests in §III
"prove tangent"	converse of tangent $\perp$ radius / alt. segment

Supporting reasons you will use constantly

radii	adj. $\angle$ s on st. line	$\angle$ s at a pt.	vert. opp. $\angle$ s	$\angle$ sum of $\triangle$	ext. $\angle$ of $\triangle$
base $\angle$ s, isos. $\triangle$	sides opp. equal $\angle$ s	alt. $\angle$ s, $AB \parallel CD$	corr. $\angle$ s, $AB \parallel CD$		
int. $\angle$ s, $AB \parallel CD$	Pyth. theorem	corr. sides, $\cong \triangle$ s	corr. $\angle$ s, $\cong \triangle$ s		
corr. sides, $\sim \triangle$ s					

Congruence: SSS, SAS, ASA, AAS, RHS. Similarity: AAA, 3 sides prop., ratio of 2 sides, inc. $\angle$.

Worked layout



AB diameter, BD tangent at B , AC produced meets BD at D , $\angle BAC = 32^\circ$. Find $\angle ADB$ and $\angle CBD$.

$$\angle ABD = 90^\circ$$

(tangent $\perp$ radius)

$$\angle ADB = 180^\circ - 90^\circ - 32^\circ = 58^\circ$$

($\angle$ sum of $\triangle$)

$$\angle CBD = \angle BAC = 32^\circ$$

($\angle$ in alt. segment)

$$\text{check: } \angle ACB = 90^\circ$$

($\angle$ in semi-circle)

Classic traps.

- Centre theorem: both angles must stand on the *same* arc, so pick the non-reflex or reflex centre angle deliberately.
- Don't use a converse to prove the theorem itself, or the theorem where the converse is needed ("prove concyclic" always needs a *converse* reason).
- Arcs are proportional to angles; **chords are not**.
- Never assume O lies on a line, a chord is a diameter, or a point is the centre unless given or proved.
- Exterior angle of a cyclic quad pairs with the *opposite* interior angle.
- Coordinate form: halve D, E after making the x^2, y^2 coefficients 1.

Beyond the reason list: prove via similar $\triangle$

Intersecting chords at P : $PA \cdot PB = PC \cdot PD$. **Tangent-secant:** $PT^2 = PA \cdot PB$.

Not standard reasons in the DSE. If you use them, prove the similar triangles first (AAA, via ($\angle$ s in the same segment) or ($\angle$ in alt. segment)), then (corr. sides, $\sim \triangle$ s).

Reason index every circle reason on one page, theorem direction vs converse

Theorem (given circle $\Rightarrow$ fact)

$\perp$ from centre bisects chord
$\perp$ bisector of chord passes through centre
equal chords, equidistant from centre
$\angle$ at centre twice $\angle$ at $\odot$ ce
$\angle$ in semi-circle
$\angle$ s in the same segment
equal chords, equal $\angle$ s
equal arcs, equal $\angle$ s
equal chords, equal arcs
arcs prop. to $\angle$ s at centre
arcs prop. to $\angle$ s at $\odot$ ce
opp. $\angle$ s, cyclic quad.
ext. $\angle$, cyclic quad.
tangent $\perp$ radius
tangent properties
$\angle$ in alt. segment

$OM \perp AB \Rightarrow AM = MB$
centre lies on it
$AB = CD \Rightarrow OM = ON$
$\angle AOB = 2\angle APB$
diameter $\Rightarrow 90^\circ$
equal angles
at centre / at $\odot$ ce
arc $\propto$ central / inscribed $\angle$
sum 180°
ext. = int. opp.
$OT \perp$ tangent
$PA = PB$, equal $\angle$ s at O and P
tangent-chord $\angle =$ alt. $\angle$

Converse (fact $\Rightarrow$ circle / tangent)

line from centre to mid-pt. of chord $\perp$ chord	$AM = MB \Rightarrow OM \perp AB$
chords equidistant from centre are equal	$OM = ON \Rightarrow AB = CD$
converse of $\angle$ in semi-circle	$90^\circ \Rightarrow$ diameter
converse of $\angle$ s in the same segment	equal angles $\Rightarrow$ concyclic
equal $\angle$ s, equal chords	
equal $\angle$ s, equal arcs	
equal arcs, equal chords	
opp. $\angle$ s supp.	sum $180^\circ \Rightarrow$ cyclic
ext. $\angle =$ int. opp. $\angle$	$\Rightarrow$ cyclic
converse of tangent $\perp$ radius	$\perp$ radius at end $\Rightarrow$ tangent
converse of $\angle$ in alt. segment	$\Rightarrow$ tangent

Wording follows the HKEAA list of acceptable abbreviations; small variants (e.g. "converse of opp. $\angle$ s, cyclic quad.") are also accepted by markers. When in doubt, a clear full sentence always earns the mark.